Smart School Food Service Analysis: AI-Driven Demand Forecasting and Waste Reduction Using Fairfax County Public Schools Data

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***Abstract*—Food waste in U.S. public schools amounts to about 530,000 tons discarded each year and $1.7 billion in losses [1]. Fairfax County Public Schools (FCPS) in Virginia aims to reduce waste using data. Although POS and kitchen records are available, most districts, including FCPS, lack tools to turn data into actionable insights. This study examines FCPS data from May 2025 to find inefficiencies, identify high- and low-demand items, and reveal patterns by region and school level. Our main goal is to provide clear, practical recommendations to help FCPS serve meals more sustainably and efficiently.**

I. INTRODUCTION

School food-service operations find themselves under increasing pressure to deliver nutritious meals at scale while reducing waste and controlling costs. As student populations grow more diverse and operational demands become more complex, school systems urgently need intelligent, data-driven solutions to optimize meal planning, procurement, and production. These efforts matter not only for efficiency, but also for child health, food security, and equity. For many children from low-income or food-insecure households, school breakfasts and lunches represent a major portion of daily caloric intake. Evidence suggests that these meals consistently outperform non-school foods in terms of diet quality. In one national sample, foods served under the U.S. federal school meal programs scored between 79–81 on the Healthy Eating Index, versus just 55–57 for non-school foods [2]. This contrast underscores that food service operations are not merely about cost savings; improving their efficiency directly supports children's health and dietary equity.

Policy-driven improvements in meal standards appear to yield both higher consumption and no increase in waste. For example, a study that compared plate waste before versus after the 2012 federal meal standard update found that students consumed 15.6% more of their entrées and 16.2% more of their vegetables after the new guidelines took effect, challenging claims that stricter nutrition rules necessarily lead to more waste [3].

Yet, despite such gains, most U.S. school districts remain without sophisticated analytics tools able to harness point-of-sale (POS), production, and waste data in real time. Without such capabilities, opportunities to further reduce waste, align procurement, tailor menus to student preferences, and monitor environmental footprint remain largely untapped.

The urgency of this problem is magnified by evidence that waste and associated inefficiencies remain substantial even in well-designed meal programs. A comprehensive review of 53 studies of the National School Lunch Program (NSLP) between 1978 and 2015 revealed wide variability in how waste is measured (direct weighing, visual estimation, photographic methods) and concluded that inconsistencies in methodology limit comparability across studies and impair understanding of long-term waste trends [4].

Beyond methodological inconsistency, more recent empirical studies continue to document sizable waste generation in school foodservice settings. For instance, a food waste audit at three U.S. schools found that food waste alone ranged from roughly 25 to 65 g per student per day and made up the largest single category of school cafeteria waste streams (ahead of milk cartons, paper, plastics, etc.) [5]. Similarly, research on school canteens in other international settings estimates that about 20–30% of prepared meals are wasted, leading to substantial financial and environmental consequences [6].

Importantly, recent intervention studies highlight that sustainable, nutritious, and cost-effective school meal programs are achievable and can even reduce environmental footprint without increasing waste or reducing student satisfaction. In one European intervention (grades 0–9), researchers applied linear optimization to redesign a four week lunch menu: the optimized menu reduced greenhouse gas emissions (GHGE) per meal by 40%, met all nutritional guidelines, cost 11% less than baseline, and yet resulted in no significant difference in food waste, student consumption, or meal satisfaction [7]. Complementary findings from sustainability assessments suggest that systematic waste-reduction strategies (such as plate-waste tracking, forecasting, pedagogic meals, and staff training) can sustainably reduce plate waste over the long term, while lowering environmental impact and avoiding nutrient loss [8].

These studies illustrate a key point: school meals are not merely about providing calories; they are a critical lever for shaping healthy, sustainable diets for children while also reducing environmental burdens and public-sector costs. And yet, as several authors note, most school food programs lack the data infrastructure and analytical capacity to implement such optimizations and monitor outcomes systematically [4].

The data provided by Fairfax County Public Schools (FCPS) covering March–May 2025 offers a rare and timely opportunity.

By combining POS, production, and waste data with modern analytic tools, including optimization modeling and real-time waste tracking, we can begin to harness this potential. With a data-driven approach, it may be possible to design menus that maximize nutritional value, minimize waste, align procurement and production with actual consumption patterns, and reduce environmental footprint, all without sacrificing meal acceptance or equity.

Thus, in the present analysis, we propose building and deploying a data-driven analytics framework for FCPS that integrates existing POS data to enable optimization and generate actionable insights for meal planning and sustainability. In doing so, we aim not only to improve operational efficiency and cost-effectiveness but to support healthier diets, food security, and environmental stewardship, advancing the broader goals of equity and sustainability in public school meal programs.

II. RELATED WORK

A. Factors Influencing Food Waste

A study conducted by Martins, Rodrigues, Cunha, and Rocha explored the factors contributing to food waste in 21 primary schools in Porto, Portugal [9]. Food waste was measured by weight and menu offerings, and students were also surveyed regarding their preferences and dietary patterns [9]. The study found that the average waste for soup was 21.6% and for main dishes was 27.5% among fourth-grade children in these primary schools [9]. Though food waste in primary schools is a global issue, where in the United States up to 49% of food is wasted, in Sweden it is 23%, in Italy it is between 20-29%, and 30-40% of fruits and vegetables were wasted in the United Kingdom [9]. The study also found that the presence of teachers in the cafeteria reduced waste; that crowded cafeterias led students to socialize rather than eat, resulting in higher waste; and that allowing students the liberty to leave the cafeteria whenever they wanted reduced food waste [9].

While optimization models address the supply side, other studies highlight the behavioral aspects of waste generation. In [10], Rada et al. conducted a comparative analysis of Italian schools, finding that waste composition shifts significantly with student age: primary schools generated more organic waste due to fruit programs, whereas high schools produced higher volumes of packaging waste from vending machines. Their findings suggest that while operational changes are critical, waste production is also heavily influenced by student and staff habits, which can be mitigated through educational 'Ecology Groups' [10]. However, unlike the present study, their approach relied on manual auditing rather than predictive modeling.

B. Cafeteria Staff Training

In addition to educational campaigns, operational strategies focused on staff training and behavioral nudges have shown efficacy. In [11], Elnakib et al. implemented the 'Smarter Lunchrooms Movement' strategies across 15 schools in a low-income district, training food service workers to improve meal presentation and lunchroom atmosphere. Their study reported a statistically significant 7.0% reduction in total food waste, with vegetable waste dropping by 7.1% and fruit waste by 13.6% [11]. Notably, they found that vegetable waste was exceptionally high, averaging 74.21% prior to intervention [11]. While these behavioral interventions are effective, they rely heavily on staff consistency in implementation, suggesting a need for automated decision-support tools to complement human efforts.

C. Implementing Mixed-Integer Linear Programming

In the broader context of food supply chain management, Mixed-Integer Linear Programming (MILP) has been successfully applied to circular economy models. The authors in [12] developed a bi-objective MILP model for a Turkish municipality to optimize the flow of food waste to reuse centers (such as food banks and animal shelters) and recycling facilities (composting and biodiesel). Their study utilized the Augmented Epsilon Constraint method to balance the conflicting objectives of minimizing network costs and maximizing waste recovery [12]. This demonstrates the versatility of MILP in handling multi-objective food waste problems. However, while their model focused on the post-consumer logistics of waste treatment, the present study applies MILP to the pre-consumer stage, aiming to optimize production planning to prevent waste generation at the source.

Beyond production planning, MILP has been widely used in downstream logistics for waste management. In [13], a MIP model was developed to solve the selective waste-collection routing problem, optimizing fleet composition and collection schedules under strict time-window constraints. The model demonstrates the capacity of integer programming to handle heterogeneous constraints—such as vehicle types and waste categories—to minimize operational costs. While the authors in [13] applied these techniques to the routing of waste-collection vehicles, the present study leverages similar MILP formulations to optimize upstream procurement and production processes, thereby addressing waste at its source.

While MILP is highly effective for discrete optimization problems like menu planning, other stages of the food supply chain often require heuristic approaches due to biological complexities. In [14], it addressed food waste prevention at the harvest stage, aiming to optimize planting schedules to avoid overproduction. The authors in [14] compared an Evolutionary Algorithm (EA) against a standard MILP formulation using the CPLEX solver. Their findings indicated that for harvest scheduling—which involves non-linear loss functions and environmental uncertainty—the evolutionary approach outperformed the exact MILP method, which struggled with the specific concave and convex objectives required [14]. This contrast highlights that while MILP is the superior tool for the cost-based, resource-constrained environment of a school cafeteria, stochastic meta-heuristics may be preferable for the biological variability found in agricultural production.

In the context of institutional catering, Akkerman, Wang, and Grunow utilized MILP to address the tactical planning of meal element distribution [15]. Their work emphasized the 'triple bottom line' of sustainability, modeling the complex trade-offs between economic efficiency, environmental impact, and product quality [15]. For instance, they demonstrated that while increasing delivery frequency preserves food quality, it simultaneously escalates transportation costs and carbon emissions. To resolve this, they proposed a weighted multi-objective function that allows decision-makers to balance financial constraints with environmental goals. This approach validates the use of MILP for managing the delicate balance between budget adherence and waste reduction in school food systems.

III. Mixed-Integer Linear Programming Optimization

Mixed-Integer Linear Programming formulates problems as linear objective functions, where, in our case, variables take integer values to address resource allocation. In [16], Moreira et al. state that MILP modeling capabilities are complemented by Python, which provides tools for formulating, solving, and visualizing models, making it accessible and a flexible resource. A primary advantage of MILP over other optimization methods is the guarantee of global optimality, ensuring that the solution found is mathematically the best possible outcome within the constraints, rather than just a local optimum [17] [18].

In [19], MILP can be formulated in the following matrix notation below:

$$
\begin{aligned}
\min_{\boldsymbol{x},\boldsymbol{y}} \quad & \boldsymbol{c}^{\mathrm{T}}\boldsymbol{x} + \boldsymbol{d}^{\mathrm{T}}\boldsymbol{y} \\
\text{s.t.} \quad & \boldsymbol{A}\boldsymbol{x} \leq \boldsymbol{b} \\
& \boldsymbol{E}\boldsymbol{y} \leq \boldsymbol{f} \\
& \boldsymbol{A} \in \mathbb{R}^{m\times n}, \boldsymbol{E} \in \mathbb{R}^{p\times q}, \\
& \boldsymbol{b} \in \mathbb{R}^{m}, \boldsymbol{f} \in \mathbb{R}^{p}, \boldsymbol{c} \in \mathbb{R}^{n}, \boldsymbol{d} \in \mathbb{R}^{q} \\
& \boldsymbol{x} \in \mathbb{Z}^{n}, \boldsymbol{y} \in \mathbb{R}^{q}
\end{aligned}
$$

The problem instance is represented by matrices and vectors $\boldsymbol{A}, \boldsymbol{E}, \boldsymbol{b}, \boldsymbol{f}, \boldsymbol{c}, \boldsymbol{d}$ from various sets of real numbers and integer numbers [19].

- $\min_{\boldsymbol{x},\boldsymbol{y}}$: Indicates the objective function, in which the goal is to minimize the result by choosing the optimal values for the decision variable vectors x and y.
- $\boldsymbol{c}^{\mathrm{T}}\boldsymbol{x}$: The dot product of cost vector c and variable vector x, where it represents the total cost associated with the integer variables.
- $\boldsymbol{d}^{\mathrm{T}}\boldsymbol{y}$: The total cost associated with the continuous variables y.
- $\boldsymbol{A}\boldsymbol{x} \leq \boldsymbol{b}$: The coefficient matrix A for the number of variables x consumes, where it has to be less than or equal to the available capacity vector b.
- $\boldsymbol{E}\boldsymbol{y} \leq \boldsymbol{f}$: The coefficient matrix E for the number of variables y consumes, where it has to be less than or equal to the limit vector f.
- $\boldsymbol{A} \in \mathbb{R}^{m\times n}$: Represents that matrix $\boldsymbol{A}$ consists of real numbers that has m constraints on n integer variables.
- $\boldsymbol{E} \in \mathbb{R}^{p\times q}$: Represents that matrix $\boldsymbol{E}$ consists of real numbers that has p constraints on q integer variables.
- $\boldsymbol{b} \in \mathbb{R}^{m}, \boldsymbol{f} \in \mathbb{R}^{p}, \boldsymbol{c} \in \mathbb{R}^{n}, \boldsymbol{d} \in \mathbb{R}^{q}$: Limit vectors b and f must have m and p real numbers respectively. The cost vectors c and d must each contain n and q real numbers, respectively.

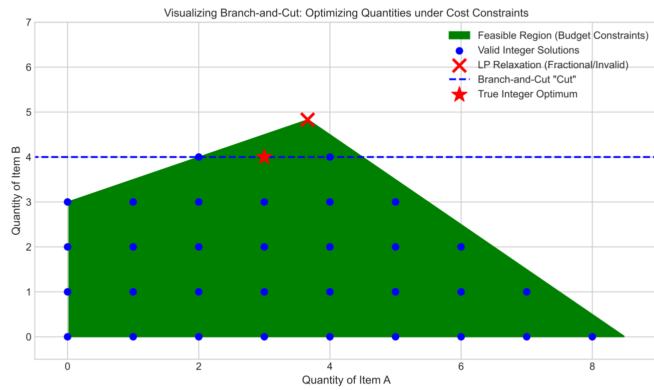

Fig. 1. Visualization of the Branch-and-Cut algorithm

- $\boldsymbol{x} \in \mathbb{Z}^{n}$: Forces every value in the vector x to be an integer number.
- $\boldsymbol{y} \in \mathbb{R}^{q}$: The values in vector y can be real numbers.

With the emergence of AI techniques to help businesses and other entities solve their problems, interpretability should not be considered. MILPs have an advantage over these AI techniques, as they are easier to understand and allow pinpointing which constraints of the MILP model led to the optimal solution [20]. In [20], interpretability will be a requirement for decision-making tools, as machine learning algorithms lack transparency and provide explanations for their results. MILPs will be useful for decision makers.

For our MILP model, we used SciPy's MILP function, which is a wrapper for the HiGHS solver [21]. Going deeper, the HiGHS Mixed Integer Programming solver is solved by a Branch-and-Cut method [21]. In [22], Genova describes the Branch-and-Cut algorithm as powerful, as it combines the Cutting Planes and Branch-and-Bound algorithms into a single framework.

To better understand the Branch-and-Cut algorithm, fig. 1 provides a small representation of the algorithm.

- **x- and y-axis**: represents the quantities of a menu item, determining how many items to produce.
- **Green shaded area**: represents the intersections that fit within the budget constraints. Where, if a point is inside the green area, it is within budget, but if the point is outside, it is either out of our budget or out of the production range. Edges are the production bounds constraints.
- **Blue points**: represent the physically possible quantities, where each point is in a whole number as needed for an ILP model.
- **Red X**: represents where a standard linear programming model would determine as the best corner of the green area. However, this fails because it falls on a fractional value; therefore, this point can never be reached.
- **Blue dashed line**: represents the cut, where the solver sees that the Red X is unachievable, so it cuts to a possible solution without losing valid points.
- **Red star**: once the cut occurs, the star represents the best integer solution for the problem.

A. Optimization Model Formulation

We created a MILP model to help FCPS cafeteria managers reduce food waste by selecting the optimal number of menu items while staying in budget. This is referred to as simultaneous optimization of control and design, where the design is our budget and the control is food production, and our model will solve both simultaneously [23]. It is also important to consider uncertainty, as mathematical models almost always contain uncertainty. Sensitivity analysis allows for different variations of the model, which can help address parameter uncertainty [24]. We implemented specific budget scenarios (80%, 100%, and 120% of the baseline budget) in our model, which provides a complete map of possible solutions as parameters change. Following the sensitivity analysis framework from [25], we tested the robustness of our model by varying key parameters. Specifically, we analyzed how the optimal production quantities changed in response to fluctuations in student demand. Since a parameter such as student demand often fluctuates, running different scenarios is ideal to maintain a safe production plan [26]. For this study, we will focus more on the baseline budget.

1) Decision Variables: Let x_j be the integer quantity of menu item j to be produced.

2) Parameters: A critical component of the MILP model is the accurate definition of cost parameters. A significant challenge in utilizing raw financial data is that school county budget documents of list expenditures as a grand total, no specified expenses that contribute to the grand total. To overcome this lack of granularity and define a usable parameter for the objective function, we derive the effective unit production cost for each item based on historical service data. The unit production cost, denoted as c_j for item j is calculated as the ratio of the total production cost to the total quality served. Utilizing the FCPS dataset provided, this is defined as:

$$c_j = \frac{\text{Production Cost Total}_j}{\text{Served Total}_j} \tag{1}$$

Where:

- Production Cost Total_j: represents the aggregate cost incurred to produce the volume of item j offered.
- Served Total_j: represents the total count of item j actually served to students (reimbursable plus non-reimbursable).

Then for the objective function, the parameters are:

- c_j: The unit production cost for item j.
- w_j: The waste penalty associated with item j (derived from historical leftover rates).
- d_j: The predicted demand for item j.

3) Objective Function: The objective is to minimize the Total Cost (C_T), balancing the financial cost of meals and the environmental cost of waste. In [27], treating environmental factors as objectives to be minimized, rather than as constraints, allows for the identification of solutions where significant environmental savings can be achieved with only marginal increases in cost. Following this logic, our model incorporates a waste penalty (λ) directly into the objective function to help reduce waste production. From [28], applying the concept of 'Value of Lost Load' to a school cafeteria setting, just as power systems assign a high financial value to preventing outages, our model assigns a penalty weight to unserved meals that exceeds the cost of food waste. This ensures that waste reduction does not come at the expense of student food security. Based on our "Minimized Cost" metric, the objective function is:

$$\text{Minimize } Z = \sum_{i=1}^{n} \sum_{j=1}^{m} (c_{ij} + \lambda w_{ij}) x_{ij} \tag{2}$$

Comparing our objective function with their function in [16], we are weighting the financial cost by λ. The waste penalty parameter λ was set to 0.1 based on the rationale that environmental costs should influence but not dominate financial optimization. However, this value was not derived from empirical estimation of actual waste disposal costs or environmental impact valuations. Different λ values could yield substantially different optimal production quantities and savings estimates. Stakeholders implementing this model should calibrate λ based on their specific waste disposal costs, sustainability priorities, and institutional values.

The waste penalty parameter (λ = 0.1) is a heuristic weight used to internalize the environmental and logistical costs of food waste. We selected λ = 0.1 as it places the model within the 'Service-Primary' stability zone. As demonstrated in our sensitivity analysis, the cost of production (including the waste penalty) does not outweigh the socio-economic penalty of an unserved meal until λ exceeds 1.50. By choosing 0.1, the optimization remains focused on maximizing student food security and dietary equity while providing a minor 'nudge' toward producing high-efficiency, low-waste menu items.

4) Constraints: Similar to the transport constraints modeled in [29], where low-volume shipments are penalized to ensure full truckloads, our model includes production and financial constraints that penalize potential waste to ensure appropriate amounts of food are produced. To align production with student needs and minimize waste, we used a production bounds constraint. This constraint keeps production within a defined range. The range is based on estimated demand:

$$L_{s,j} \leq x_{s,j} \leq U_{s,j} \quad \forall s, \forall j \tag{3}$$

In our optimization model, we scale the bounds by size multipliers tied to each school's student population (Table I).

TABLE I
PRODUCTION BOUNDS BASED ON SCHOOL SIZE

Student Population	**Production Bounds** (L,U)
0–499	(0.80, 1.20)
500–999	(0.85, 1.15)
1000–1499	(0.90, 1.10)
1500–1999	(0.95, 1.05)
2000–2499	(0.96, 1.04)
2500–2999	(0.97, 1.03)
3000–3499	(0.98, 1.02)
3500+	(0.99, 1.01)

We see that smaller schools have a longer range than the larger schools. Schools with smaller populations are more volatile, as a small change in daily events can result in a significant impact on the waste percentage. For example, a school of 500 students has a flu outbreak one day, and 50 students call in sick; that is a 10% reduction in demand that day. A wider buffer accounts for this unpredictability. On the other hand, a school of 3000 students with 50 students calling out on any given day would result in a 1.6% reduction in demand. The buffer would be tighter as attendance is more stable. This is represented by:

$$L_{s,j} = \alpha_s \cdot d_{s,j}$$
$$U_{s,j} = \beta_s \cdot d_{s,j} \tag{4}$$

Where coefficients (α_s, β_s) are determined by school size:

$$(\alpha_s, \beta_s) = \begin{cases} (0.80, 1.20) & \text{if } s \in \text{0-499} \\ (0.85, 1.15) & \text{if } s \in \text{500-999} \\ (0.90, 1.10) & \text{if } s \in \text{1000-1499} \\ (0.95, 1.05) & \text{if } s \in \text{1500-1999} \\ (0.96, 1.04) & \text{if } s \in \text{2000-2499} \\ (0.97, 1.03) & \text{if } s \in \text{2500-2999} \\ (0.98, 1.02) & \text{if } s \in \text{3000-3499} \\ (0.99, 1.01) & \text{if } s \in \text{3500+} \end{cases}$$

Here, $d_{s,j}$ represents the demand for item j at school s. In addition to these production bounds, we require that the production quantities, represented by the decision variables, be non-negative integers, enforced by:

$$x_{s,j} \in \mathbb{Z}^+ \quad \forall s, \forall j \tag{5}$$

It is important to know, however, that using integer variables creates a discontinuous solution space. When a parameter changes, the optimal solution often performs jumps throughout the space [24]. If a small budget change were to occur in the county, our optimal food recommendations would change drastically, as expensive foods could be dropped entirely, allowing cheaper foods to fill the allocated space.

Finally, we included two budget constraints in the model: a per-school budget constraint (B_s), and a system-wide total budget constraint (B_{total}). This is considered a resource constraint, as, in our case, financial capital, the budget needs to stay within the range of what it is allocated to [16]:

$$\sum_j (c_j \cdot x_{s,j}) \leq B_s \tag{6}$$

- c_j: The unit production cost for item j.
- $x_{s,j}$: Quantity of items produced by school s.
- B_s: The maximum budget allocated to school s.

The per-school budget constraint ensures equity across all the schools in the system. It prevents larger schools from "hoarding" the budget, ensuring every school has enough funds to meet its specific needs.

$$\sum_s \sum_j (c_j \cdot x_{s,j}) \leq B_{total} \tag{7}$$

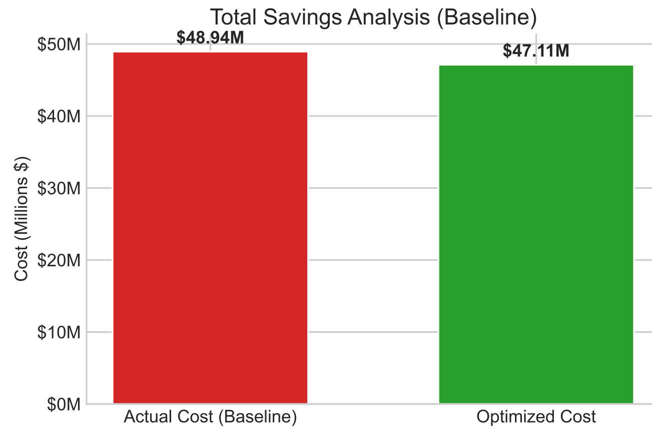

Fig. 2. Overall Savings Analysis. This comparison shows the financial impact of the optimization model across the system. The red bar shows the estimated actual annual baseline cost ($48.94M), and the green bar shows the lower cost reached with the optimized menu planning model ($47.11M).

The system-wide budget constraint ensures sustainability. It ensures that the spending across the county does not exceed the allocated $139,144,760 budget [30]. Implementing a system-wide budget constraint is ideal, as the entire school system must remain within the total budget while optimizing individual schools. This allows the model to optimize the allocation of funds between purchasing higher-quality (less wasteful) ingredients and the labor needed to prepare them, ensuring that the solution remains financially feasible [31].

IV. RESULTS

A. Financial Impact Analysis

Fig. 2 shows a comparison of production costs between actual realized costs from the data and our model's estimated annual cost. We found that the actual annual cost for the entire school system in May 2025 was $48.94 million. After adjusting production and budgetary limits, our model estimated an annual cost of $47.11 million. The projected annual savings of approximately $1.83 million represent a theoretical upper bound under several key assumptions: (1) that May production patterns are representative of the full school year, (2) that reduced production would not affect student purchasing behavior, (3) that historical sales accurately reflect student demand without supply constraints, and (4) that the assumed budget allocation by school size reflects actual practice. While the optimization model is mathematically sound, these projections should be treated as indicative of potential savings magnitude rather than as precise forecasts. We recommend pilot implementation at a subset of schools to validate these estimates before system-wide deployment.

The projected cost savings assume that reducing production quantities would proportionally reduce waste while maintaining the same sales volume. This counterfactual assumption may not hold if student purchasing behavior responds to perceived availability, menu variety, or presentation factors that change with production volumes. The optimization results represent potential savings under idealized conditions rather than guaranteed outcomes. Prospective validation through controlled

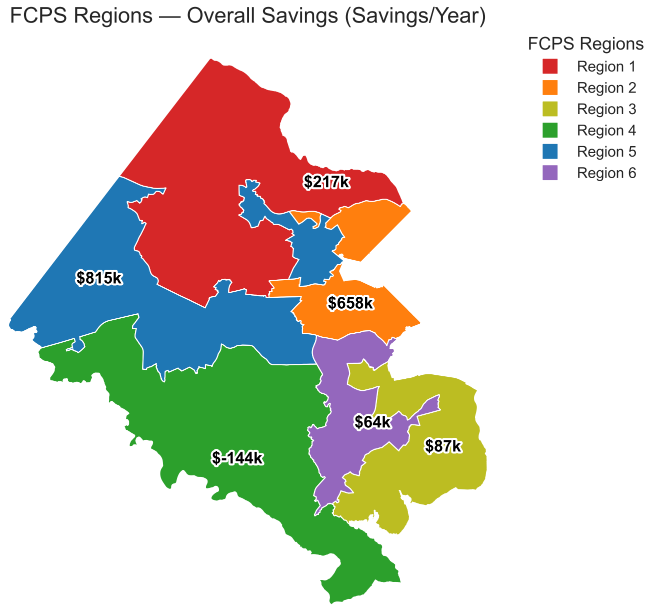

Fig. 3. Choropleth map of Fairfax County with FCPS regions, where the colors represent the different regions and are labeled by the amount of savings/loss each region has generated by the optimization

implementation at selected schools would be necessary to confirm these projections.

According to the National Center for Educational Statistics [32], as of 2022, Fairfax County Public Schools serves:

- **Total Population**: 1,145,354
- **Median Household Income**: $145,165
- **Total Households**: 410,844
- **Race/Ethnicity**: 49% White, 20% Asian, 17% Hispanic/Latino, 10% African American, 5% Other
- **Households with Internet**: 95.9%

In terms of the FCPS regions, here are key points to describe the savings amounts from fig. 3:

- **Region 1**: *Average Median Household Income*: $170,330 [33]
  The two Census districts of Dranesville and Hunter Mill make up this region and have a median household income of $199,209 [34] and $164,050 [35] for 2023, respectively. The northern parts of the region are the wealthiest in the county, as cities like Great Falls and McLean are located there. Therefore, students in the northern parts of the region have the option to bring their own lunch, which creates demand volatility. However, the data show that the region had better historical management of production costs, leading to moderate savings across the whole county.
- **Region 2**: *Average Median Household Income*: $123,007 [33]
  In [36], a couple of areas in the south-eastern parts of the region are experiencing the greatest need for food, whereas the northern part is the complete opposite. The region's northern cities include Tyson's Corner and some neighborhoods of McLean and Langley. The mix of rich and poor areas divides the food production needs of these school cafeterias.
- **Region 3**: *Average Median Household Income*: $134,059 [33]
  According to Fairfax County's Needs Assessment of 2022 [36], 11 out of the 18 Census tracts in Fairfax County ranked as the greatest need for food reside in this region. This would mean that students in these schools experience food insecurity and benefit from the food provided by the schools. Therefore, savings in this area are low, as the food production required meets demand in this region.
- **Region 4**: *Average Median Household Income*: $176,789 [33]
  In [37], this region has the lowest savings due to the location of the state's largest schools, Lake Braddock Secondary and Robinson Secondary. The student populations at Lake Braddock Secondary and Robinson Secondary are 5422 and 4904, respectively [38]. These schools have massive student populations, and so they serve far more meals than smaller high schools in other regions. Therefore, this region produces the highest volume of waste due to its student population.
- **Region 5**: *Average Median Household Income*: $164,789 [33]
  Being at the heart of Fairfax County, this region has generated the highest amount of savings from our optimization model. Students in this category come from relatively high-income households, allowing their parents to afford lunch for their children. This causes demand volatility, as it is unpredictable to produce an optimal amount of food for the schools in advance.
- **Region 6**: *Average Median Household Income*: $148,114 [33]
  Schools like Annandale High and Lewis High have high student participation of in meal programs, which means

TABLE II
SENSITIVITY ANALYSIS OF WASTE PENALTY (λ) ON PRODUCTION STRATEGY

λ Range	Total Monthly Units	Strategy Profile	Service Priority
0.01 – 1.49 (Selected: 0.1)	**12,931,708**	**Service-Optimized**	**High (115%)**
1.50 – 1.90	12,360,530	Transitional	Moderate
1.91 – 2.00	9,558,220	Waste-Conservative	Low (85%)

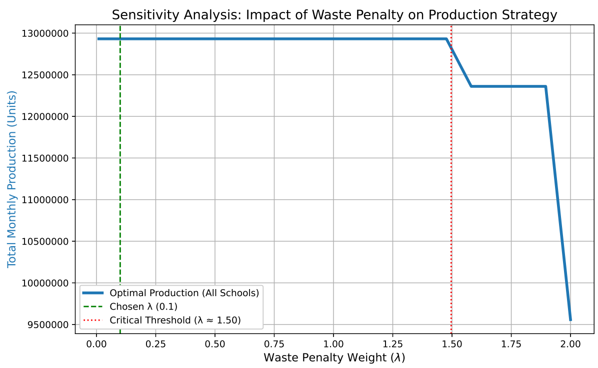

Fig. 4. Sensitivity Analysis of the Waste Penalty. The plot illustrates the robustness of production strategy across a range of λ values (0.01 to 2.0). Total production remains stable at the upper bound (115% of demand) for $\lambda \leq 1.50$, prioritizing service assurance. The chosen value of λ is situated well within this stability zone. A transition toward a waste-conservative strategy occurs only after the critical threshold of $\lambda \approx 1.50$.

that they are in high need. This results in schools producing the amount of food that meets the demand of students. This region also is home to Hayfield Secondary School, where it has a high student population. Given the size, demand volatility is introduced as demand is unpredictable as students have the choice to pack or buy lunch. Altogether, this is a stabilized region as it lacks the extreme peaks of wealth that drive unpredictability.

Fig. 5 shows the financial performance of the optimization model. The scatter plot compares each school's actual annual food costs (x-axis) with its optimized annual costs (y-axis). The dashed diagonal line marks the point where optimized costs equal historical spending.

Schools below the line, shown in green, are where the model found inefficiencies and projected cost savings. The model placed 130 schools in this group. In contrast, 60 schools above the line, shown in red, are expected to have higher costs. This likely happens when past procurement did not meet the model's stricter rules. The size of each marker shows the financial impact.

There is a clear correlation between school size and optimization results. The model saves the most money in schools with 500 to 1,500 students. Schools with more than 2,000 students resulted in higher costs and negative savings (see Fig. 6). Breaking down the financial data, smaller and medium-sized schools used to overproduce, which the model fixed. In contrast, larger schools may not have produced enough for their students. The model increased food allocations for these schools to ensure students are properly served.

B. Interpretation of Projected Cost Increases

The optimization model projected a cost increase for a small group of schools, as indicated by data points above the break-even line in Fig. 5. While reverting these schools to historical production patterns may appear financially prudent, this perspective fails to account for the model's embedded constraint logic.

The optimization strictly enforces demand satisfaction constraints ($x_{s,j} \geq L_{s,j}$). This ensures production matches estimated student consumption. The historical "savings" in these schools likely result from under-production, when supply did not meet demand. This may have led to items running out or reduced service levels. The model's projected cost increase should not be seen as inefficiency, but as the *true cost of service assurance*. For these schools, the model shows a gap between past budgeting and the real resources needed to fully serve students.

C. Sensitivity Analysis of Waste Penalty (λ)

To validate the empirical robustness of our chosen parameter, we conducted a sensitivity analysis by varying λ from 0.01 to 2.0. The results (see fig. 4 and Table II) reveal a significant stability zone between 0.01 and 1.50. Within this range, the total monthly production quantity remains constant at 12.93 million units (representing 115% upper bound of demand). This confirms that for values of λ up to 1.50, the 'cost' of food waste is mathematically secondary to the mission of service assurance. A critical threshold is reached at $\lambda \approx 1.50$, where the model begins to shift toward a 'Waste-Conservative' strategy, reducing production to minimize the financial penalty of leftovers. Our selection of $\lambda = 0.1$ ensures the system is resilient to data fluctuations and consistently prioritizes the nutritional needs of the Fairfax County student population.

V. REGRESSION ANALYSIS

To further investigating the factor of the school food service system, we conducted a regression analysis to predict the production cost. This analysis serves as an additional tool to validate cost behaviors descriptively.

A. Methodology: Ordinary Least Squares

We utilized the Ordinary Least Squares (OLS) regression to estimate the relationship between a dependent variable(Y) and one or more independent variables (X) by minimizing the sum of the squared differences (residuals) between the observed values and the values predicted by the linear model [39].

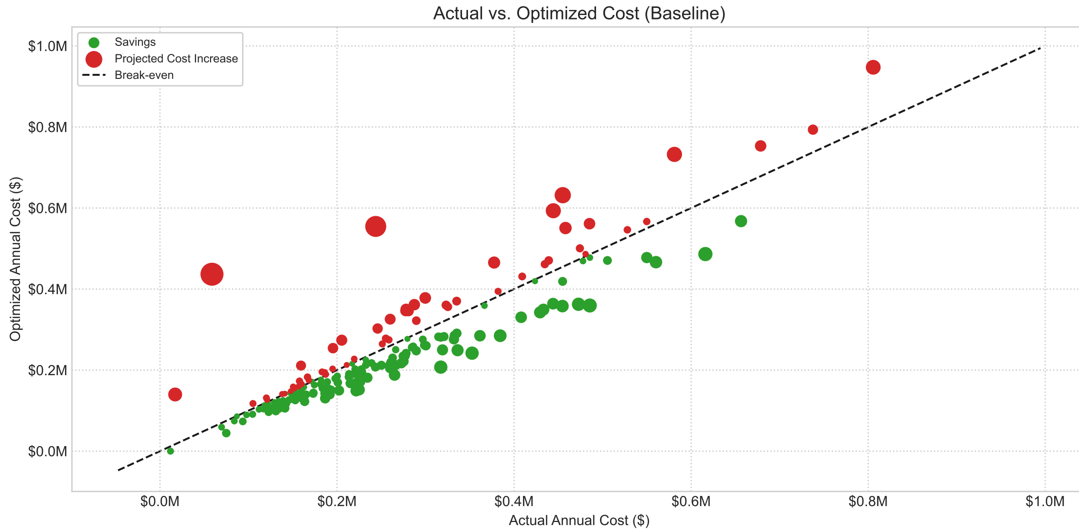

Fig. 5. Savings Analysis: Actual vs. Optimized Cost. Green circles represent schools achieving savings, while red circles indicate projected losses. The size of the circle corresponds to the magnitude of the financial impact.

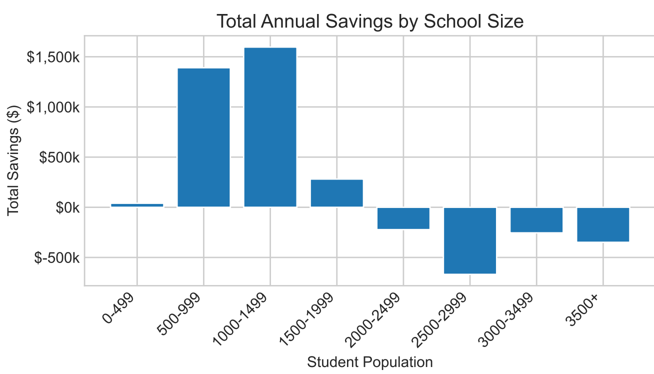

Fig. 6. Total Annual Savings by School Size Category. The chart shows that, by student population, the optimization model usually leads to net savings (positive values), but in some cases results in higher costs when demand constraints must be met (negative values).

The general linear model is defined as [40]:

$$Y_i = \beta_0 + \sum_{j=1}^{k} \beta_j X_{ij} + \varepsilon_i \tag{8}$$

Where:

- Y_i represents the dependent variable (e.g., Production Cost) for school i.
- β_0 is the intercept, representing fixed operational baselines.
- β_j represents the marginal impact of the j-th independent variable.
- X_{ij} represents the independent variables (e.g., student enrollment, waste metrics).
- ε_i is the error term, assumed to be normally distributed $\varepsilon \sim N(0, \sigma^2)$.

B. *Objective Function and Implementation*

The primary goal of OLS method is to determine the optimal values for the coefficients $\hat{\beta}_0$ and $\hat{\beta}_1$ that minimize the Sum of Squared Residuals (SSR). A lower SSR value indicates a better-fitting model that makes more accurate predictions. The objective function is mathematically formulated as [41]:

$$\min_{\beta_0,\beta_1} S(\beta_0, \beta_1) = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 = \sum_{i=1}^{n} (y_i - (\beta_0 + \beta_1 x_i))^2 \tag{9}$$

To validate the model's accuracy and reliability on predicting production cost, the methodology strictly incorporates a cross-validation protocol. The dataset is split into a training set (80%) and a test set (20%).The algorithm exclusively utilized the training set to derive the coefficients. Then these learned patterns are applied to the test set to generate the prediction model. Finally, the predictions are compared with the original production cost in test set to quantify how accurate and how reliable the model is. By minimizing this function, we ensure that the predicted production costs ($\hat{y}$) are as close as possible to the historical data provided by FCPS.

C. *Future Pathway*

Although the regression component operates independently of the MILP optimization presented in Section III, it provides essential statistical validation for the cost parameters utilized in the optimization constraints. Currently this regression framework is *descriptive*,providing a statistical explanation of historical costs. In our future research, we can use our regression outputs inside the optimization models to explore decisions or predictions under different constraints [42].

The coefficients derived from the OLS analysis (β_j)serve as the objective function parameters for a Linear Programming (LP)problem. For example, once the marginal cost of waste (β_{waste}) is empirically determined, an optimization solver can be formulated to minimize total operational cost Z:

$$\text{Minimize } Z = \sum_{i=1}^{n} (\beta_{prod} \cdot x_{i,prod} + \beta_{waste} \cdot x_{i,waste}) \tag{10}$$

TABLE III
TOP 10 MOST WASTED BREAKFAST PRODUCTS

Product Name	Leftover Rate (%)
Fat Free Chocolate Milk(Each)	98.8
Cereal, Cinnamon Toasters, Malt-O-Meal, IW	60.3
Fat Free White Milk(Each)	53.5
Cherry Flavored Dried Cranberries(Each)	53.1
Fat Free Unflavored Shelf-Stable Milk(Each)	52.7
Milk, Lactose Free, Fat Free, Shelf-Stable(Each)	51.3
Orange Flavored Dried Cranberries(Each)	50.0
Watermelon Flavored Dried Cranberries(2 Eaches)	48.0
Assorted Cereal(Cereal Cup)	47.1
Patty, Chicken, Breaded, WG(2 strips)	46.9

TABLE IV
TOP 10 MOST WASTED LUNCH PRODUCTS

Product Name	Leftover Rate (%)
Fat Free White Milk(Each)	60.6
Milk, Lactose Free, Fat Free, Shelf-Stable(Each)	50.9
1% White Milk(Each)	50.8
Fat Free Unflavored Shelf-Stable Milk(Each)	48.8
Turkey, Breast, Pre-Sliced, Oven Roasted(6 Slices)	47.1
String Cheese(Each)	42.8
1% Unflavored Shelf-Stable Milk(Each)	39.9
Assorted 4 oz Yogurt(Container)	39.0
Mustard Packet(Packet)	34.7
Lactose Free, White Milk(Each)	32.9

Subject to demand and capacity constraints:

$$x_{i,prod} \geq D_i \quad \forall i \in \text{Schools} \tag{11}$$

$$x_{i,prod} \leq C_{cap} \tag{12}$$

By feeding the empirically derived regression coefficients into the optimization model, the system evolves from merely identifying inefficiencies to algorithmically prescribing production quotas that minimize waste and maximize budgetary efficiency.

VI. CHALLENGES AND LIMITATIONS

A. Challenges

An issue that other researchers observed was student acceptance of certain foods. In Mulhollem's article [43], they observed that students, particularly elementary school students, avoided certain foods that were served to them. "Because they are the most highly wasted, understanding the extent to which fruit and vegetable offerings in school lunches are likely to be accepted by children has important implications for school meal policies and children's health" [43]. On the other side, "eggs and poultry, the researchers found, were the least-wasted food category" [43]. According to our findings, milk products are the most discarded items in both Table III and Table IV. FCPS should not worry that students are avoiding their fruits and vegetables, though cranberry products are among the top 10 breakfast items wasted (Table III). Students eating in FCPS schools are eating relatively healthy. However, with milk items frequently being wasted, this points to a problem with their waste management.

In their study of the private school in Columbia, Missouri, Mulhollem found that changing the cafeteria menu can be difficult [43]. "Addressing menu changes while staying within budget and offering children nutritious meals often is not an easy task" [43]. In the 2025 school year, Fairfax County Public Schools (FCPS) introduced new menu items [44], as shown in Table V. Since students may not be familiar with these options, there is a real risk of increased food waste if they choose not to eat them. This could worsen current waste management challenges, especially given the high production costs and large quantities involved, which may result in financial losses. Because these items are new, it will be important to conduct a follow-up study to assess their effectiveness and their impact on waste management.

TABLE V
NEW SCHOOL MENU ITEMS FOR THE CURRENT SCHOOL YEAR

New Product Name	**Category**
Cinnamon Apple Overnight Oats	Breakfast
Egg & Cheese Breakfast Quesadilla	Breakfast
Egg & Cheese Breakfast Taco	Breakfast
French Toast Sticks	Breakfast
Egg & Cheese Bagel	Breakfast
Turkey Sausage & Cheese Bagel	Breakfast
Sofritas	Lunch
Peruvian Chicken Drumstick	Lunch
Spanish Rice	Lunch
Cheesy Turkey Pepperoni Pasta	Lunch
Cheesy Pasta with Marinara Sauce	Lunch
Tomato Grilled Cheese	Lunch
Deep Dish Pizza	Lunch
Chicken Masala Pizza	Lunch
Macaroni & Cheese	Lunch
Brunch for Lunch (Cheesy Scrambled Eggs, Potato Wedges, French Toast Sticks, Turkey Sausage)	Lunch
Veggie Chili	Lunch
Patty Melt	Lunch
Buffalo Chicken Dip	Lunch
Teriyaki Chicken or Chickenless Bites Stir-Fry with Fried Rice and Asian Stir-Fry Vegetables	Lunch

Another problem brought up is the reliance on a single individual to sustain a waste management program. Huber's article [45] describes how one teacher, Gerin Hennebaul, initiated the push for change in waste management with her students, which subsequently led to waste-management initiatives. However, if she were the only advocate for this cause and left the school, the program could collapse. In Fairfax County Public Schools, the administration is actively seeking assistance to reduce waste. This approach would require a collective effort from administrative staff and encourage passionate individuals within schools to participate. Though FCPS may be less vulnerable to this issue, it is important that the administration is committed to promoting waste management across the entire school system to ensure widespread adoption.

B. Limitations

A significant limitation of this study is the reliance on production data from a single month (May 2025). May represents an atypical period in the school calendar, characterized by end-of-year activities, standardized testing schedules, senior events, and generally declining attendance patterns. Extrapolating these findings to annual estimates assumes that May consumption patterns are representative of the full academic year, which may not hold. Seasonal variation in food preferences, weather-dependent consumption changes, and academic calendar effects could substantially alter optimal production quantities in other months. The reported annual savings of $1.83 million should therefore be interpreted as an estimate conditional on May patterns persisting year-round, rather than a validated projection.

Our model uses historical sales data as a proxy for student demand; however, this approach carries an inherent limitation. If popular items historically ran out before all interested students could purchase them, observed sales would underestimate true demand. In such cases, our optimization model might recommend further reductions in production, potentially exacerbating supply shortfalls rather than reducing waste. Without explicit data on unmet demand or stockout events, we cannot distinguish between low demand and supply-constrained sales. Future implementations should incorporate stockout tracking to refine demand estimates.

Our analysis assumes that school budgets are allocated proportionally to student population rather than equally across schools. Additionally, the Food and Nutrition Services budget category labeled 'Expenditures' was assumed to primarily fund food production, though it likely includes personnel costs, equipment, maintenance, and compliance expenses. These assumptions were necessary given available budget documentation but may not reflect actual allocation practices. The magnitude of projected savings could differ substantially under alternative budget allocation models, and future work with access to detailed line-item budgets would improve estimate precision.

VII. CONCLUSION

This study tackled economic and environmental food waste issues in Fairfax County Public Schools using data analysis and an integer linear programming optimization model. The model projected an annual system-wide cost reduction of about $1.83 million, lowering spending from $48.94 million to $47.11 million. This shows the system can reclaim significant economic value without lowering nutritional standards. The graphs show that the impact of optimization varies by school size. Schools with 500–1500 students had the greatest total savings, while larger schools faced a projected cost increase. This increase does not indicate inefficiency; it reveals historical under-production. The model corrects this to ensure resources fully meet student demand.

We acknowledge that the practical realization of these savings requires validation beyond the scope of this study. The single-month data window, uncertainty in budget allocation, and untested behavioral assumptions collectively suggest that our estimates should be interpreted as demonstrating the feasibility and potential magnitude of optimization-driven improvements rather than as guaranteed outcomes. We recommend that FCPS consider a phased implementation approach: first deploying the model at 5-10 pilot schools across different regions and size categories, measuring actual waste reduction and cost impacts over a full semester, and refining model parameters based on observed outcomes before system-wide adoption.

Future iterations of this study could incorporate additional environmental metrics beyond food waste, such as reducing a school's carbon footprint or water usage. To manage something like this, MILP-based clustering methods could be deployed to facilitate decision-making on environmental objectives for school administrators [27]. While our current model uses deterministic constraints, future work could adopt fuzzy inventory modeling techniques, as explored by [46], to better account for the vagueness and uncertainty in student demand and production costs.

The regression analysis presented in Section V currently operates independently of the MILP optimization and serves a descriptive rather than prescriptive function. While the OLS framework provides statistical validation of cost relationships, the connection between regression outputs and optimization parameters remains conceptual rather than implemented. Future work should formally integrate these components, using regression-derived coefficients directly within the optimization objective function as outlined in Section V.C

VIII. DECLARATIONS

Data Availability

The datasets used and analyzed during the current study are publicly available in the "fall-2025-group2 repository", accessible at: https://github.com/JupiK9/fall-2025-group2.git. The repository includes the raw data, processed data, and code necessary to reproduce the results presented in this study.

Conflict of Interest

The authors declare that they have no competing interests.

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